



Province of the
EASTERN CAPE
EDUCATION

Iphondo leMpuma Kapa: Isebe leMfundo
Provinsie van die Oos Kaap: Departement van Onderwys
Porafensie Ya Kapa Botjhabela: Lefapha la Thuto

**NATIONAL
SENIOR CERTIFICATE/
NASIONALE
SENIOR SERTIFIKAAT**

GRADE/GRAAD 11

NOVEMBER 2025

**TECHNICAL MATHEMATICS P1/
TEGNIJSE WISKUNDE V1
MARKING GUIDELINE/NASIENRIGLYN**

MARKS/ PUNTE: 150

MARKING CODES/NASIENKODES	
A	Accuracy/Akkuraatheid
CA	Consistent accuracy/Volgehoue akkuraatheid
M	Method/Metode
R	Rounding/Afronding
NPR	No penalty for rounding/Geen penalisering vir afronding nie
NPU	No penalty for units omitted /Geen penalisering vir eenhede weggelaat nie
S	Simplification/Vereenvoudiging
SF	Substitution in correct formula/Vervanging in korrekte formule

This marking guideline consists of 15 pages/
Hierdie nasienriglyn bestaan uit 15 bladsye

QUESTION / VRAAG 1		
1.1.1	$\frac{3a^0}{2} + \left(\frac{b}{3a}\right)^0$ $= \frac{3}{2} + 1$ $= \frac{5}{2}$	$\checkmark \frac{3}{2}$ A $\checkmark 1$ A $\checkmark \frac{5}{2}$ A (3)
1.1.2	$\left(\frac{1}{81}\right)^{-\frac{3}{4}}$ $= (3^{-4})^{-\frac{3}{4}}$ $= 27$	$\checkmark 3^{-4}$ A $\checkmark 27$ CA (2)
1.1.3	$\sqrt[3]{125x^6} - \sqrt[4]{81x^8} + (36x^4)^{\frac{1}{2}}$ $= \sqrt[3]{5^3x^6} - \sqrt[4]{3^4x^8} + (6^2x^4)^{\frac{1}{2}}$ $= 5x^2 - 3x^2 + 6x^2$ $= 8x^2$	$\checkmark 5x^2$ A $\checkmark -3x^2$ A $\checkmark 6x^2$ A $\checkmark 8x^2$ CA (4)
1.1.4	$\frac{18^{2n+1} \cdot 9^{-2n} \cdot 16^{n-1}}{64^{n+1}}$ $= \frac{(2 \times 3^2)^{2n+1} \cdot (3^2)^{-2n} \cdot (2^4)^{n-1}}{(2^6)^{n+1}}$ $= \frac{2^{2n+1} \cdot 3^{4n+2} \cdot 3^{-4n} \cdot 2^{4n-4}}{2^{6n+6}}$ $= 2^{2n+1+4n-4-6n-6} \cdot 3^{4n+2-4n}$ $= 2^{-9} \cdot 3^2$ $= \frac{9}{512}$	$\checkmark 2^{2n+1} \cdot 3^{4n+2}$ A $\checkmark 3^{-4n}$ A $\checkmark 2^{4n-4}$ A $\checkmark 2^{6n+6}$ A $\checkmark 2^{-9} \cdot 3^2$ CA $\checkmark \frac{9}{512}$ CA (6)

1.1.5	$\frac{3(x^4 y^{-4})^2}{2xy^2} \div \frac{(xy)^{-3}}{(3x^{-3} y^4)^2}$ $= \frac{3x^8 y^{-8}}{2xy^2} \times \frac{9x^{-6} y^8}{x^{-3} y^{-3}}$ $= \frac{27}{2} x^4 y$	$\checkmark 3x^8 y^{-8}$ $\checkmark \mathbf{M}$ $\checkmark 3x^{-6} y^8$ $\checkmark x^{-3} y^{-3}$ $\checkmark \frac{27}{2} x^4 y$	A A A A CA (5)
1.2.1	$\frac{\sqrt{1000}}{\sqrt{2}(\sqrt{80} - \sqrt{45})} = 10$ $\text{L.H.S} = \frac{\sqrt{10 \times 100}}{\sqrt{2}(\sqrt{16 \times 5} - \sqrt{9 \times 5})}$ $= \frac{10\sqrt{10}}{\sqrt{2}(4\sqrt{5} - 3\sqrt{5})}$ $= \frac{10\sqrt{2}\sqrt{5}}{\sqrt{2}\sqrt{5}} \quad \text{or} \quad \frac{10\sqrt{10}}{\sqrt{10}}$ $= 10$ $\text{L.H.S} = \text{R.H.S}$	$\checkmark \sqrt{10 \times 100}$ $\checkmark \sqrt{16 \times 5}$ $\checkmark \sqrt{9 \times 5}$ $\checkmark \frac{10\sqrt{2}\sqrt{5}}{\sqrt{2}\sqrt{5}} \quad \text{or} \quad \frac{10\sqrt{10}}{\sqrt{10}}$	A A A A (4)

1.2.2	$\log_5 125 - \frac{\log 32 - \log 8}{\log 16} = \frac{5}{2}$ L.H.S $= \log_5 5^3 - \frac{\log 2^5 - \log 2^3}{\log 2^4}$ $= 3 \log_5 5 - \frac{5 \log 2 - 3 \log 2}{4 \log 2}$ $= 3(1) - \frac{2 \log 2}{4 \log 2}$ $= 3 - \frac{1}{2}$ $= \frac{5}{2}$ L.H.S = R.H.S OR/OF $= \log_5 5^3 - \frac{\log \frac{32}{8}}{\log 16}$ $= 3 \log_5 5 - \frac{\log 4}{2 \log 4} \text{ or } 3 \log_5 5 - \frac{2 \log 2}{4 \log 2}$ $= 3(1) - \frac{1}{2}$ $= \frac{5}{2}$	✓ rewrite with prime bases / <i>skryf as priembasisse</i> M A ✓ power property / <i>magseienskap</i> A ✓ S CA ✓ 3 A ✓ $\frac{1}{2}$ A OR/OF ✓ quotient law / <i>kwosiëntwet</i> A ✓ power property / <i>magseienskap</i> A ✓ S CA ✓ 3 A ✓ $\frac{1}{2}$ A (5)														
1.3.1	$A + B$ $= 75 + 15$ $= 90$	✓ Substitution / <i>vervanging</i> A ✓ 90 A AO: Full marks / volpunte (2)														
1.3.2	<table><tr><td>64</td><td>32</td><td>16</td><td>8</td><td>4</td><td>2</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td></tr></table>	64	32	16	8	4	2	1	1	0	1	1	0	1	0	✓ M A ✓ 1011010₂ A AO: Full marks / volpunte (2)
64	32	16	8	4	2	1										
1	0	1	1	0	1	0										
1.4.1	$V = l \times b \times h$ $V = 60 \times 30 \times 25$ $V = 45\,000\text{ cm}^3$	✓ SF A ✓ 45 000 CA (2)														

1.4.2	$4,5 \times 10^4$	$\checkmark 4,5 \times 10^4$	A (1)
			[36]

QUESTION / VRAAG 2		
2.1.1	$0,5x^3 = 0,0625$ $x^3 = \frac{1}{8}$ $x = \left(\frac{1}{2^3}\right)^{\frac{1}{3}}$ $x = \frac{1}{2}$	$\checkmark x^3 = \frac{1}{8}$ A $\checkmark \left(\frac{1}{2^3}\right)^{\frac{1}{3}}$ CA $\checkmark x = \frac{1}{2}$ CA (3)
2.1.2	$3^x + 3^{x+1} = 36$ $3^x + 3^x \cdot 3^1 = 36$ $3^x(1+3) = 36$ $3^x = 9$ $3^x = 3^2$ $x = 2$	$\checkmark 3^x \cdot 3^1$ A $\checkmark$ Factorization / <i>faktoriseer</i> A $\checkmark 3^x = 3^2$ CA $\checkmark x = 2$ CA (4)
2.1.3	$(-2\sqrt{3+x})^2 = 12$ $4(3+x) = 12$ $12 + 4x = 12$ $4x = 0$ $x = 0$	$\checkmark 4(3+x)$ A $\checkmark$ Simplification / <i>vereenvoudiging</i> CA $\checkmark x = 0$ CA (3)
2.1.4	$\log_3(x+4) + \log_3(x-4) = \log 100$ $\log_3(x+4)(x-4) = \log 10^2$ $\log_3(x^2 - 16) = 2$ $x^2 - 16 = 3^2$ $x^2 - 25 = 0$ $(x-5)(x+5) = 0$ $x = 5$ or $x = -5$ $\therefore x = 5$	$\checkmark$ log property of addition / <i>log optellingswet</i> A $\checkmark 2\log 10$ A $\checkmark \log_3(x^2 - 16) = 2$ CA $\checkmark$ log property / <i>log eienskap</i> CA $\checkmark$ Factors / <i>faktore</i> CA $\checkmark x$ value / <i>waarde</i> CA (6)
2.2.1	$P = \frac{V^2}{R}$ $V^2 = RP$ $V = \sqrt{RP}$	$\checkmark V^2$ subject / <i>onderwerp</i> A $\checkmark V = \sqrt{RP}$ CA (2)

2.2.2	$V = \sqrt{25 \times 100}$ $V = 50$ volts	✓ SF ✓ $V = 50$ volts	A CA (2)
			[20]

QUESTION / VRAAG 3			
3.1.1	$x(x-1) = 0$ $x = 0$ or $x = 1$	✓ 1 ✓ 0	A A (2)
3.1.2	$x^2 - x = 1$ $x^2 - x - 1 = 0$ $x = \frac{-(-1) \pm \sqrt{(-1)^2 - 4(1)(-1)}}{2(1)}$ $x = \frac{1 \pm \sqrt{5}}{2}$ $x = 1,62$ or $x = -0,62$	✓ Standard form / <i>standaardvorm</i> ✓ Substitution / <i>vervang</i> ✓✓ Values of/ <i>waardes van x</i>	A CA CA (4)
3.1.3	$(x-3)(x-4) \geq 0$ Critical values / <i>Kritiese waardes</i> $x = 3$ or $x = 4$ $x \leq 3$ or $x \geq 4$ OR $x \in (-\infty; 3] \cup [4; \infty)$	✓ Both critical values / <i>beide kritiese waardes</i> ✓ $x \leq 3$ ✓ $x \geq 4$	A CA CA (3)
3.2	$2y + x = 5$ and $2y^2 - xy - 4x^2 = 8$ $x = 5 - 2y \dots\dots\dots(3)$ $2y^2 - y(5 - 2y) - 4(5 - 2y)^2 = 8$ $2y^2 - 5y + 2y^2 - 4(25 - 20y + 4y^2) = 8$ $2y^2 - 5y + 2y^2 - 100 + 80y - 16y^2 - 8 = 0$ $-12y^2 + 75y - 108 = 0$ $y = \frac{-(75) \pm \sqrt{(75)^2 - 4(-12)(-108)}}{2(-12)}$ $y = \frac{9}{4}$ or $y = 4$ $x = 5 - 2\left(\frac{9}{4}\right)$ or $x = 5 - 2(4)$ $x = \frac{1}{2}$ or $x = -3$	✓ Equation/ <i>vergelyking</i> 3 ✓ Substitution/ <i>vervang</i> ✓ Standard form/ <i>standaardvorm</i> ✓ Factors / <i>faktore</i> ✓ Both values of <i>y/beide waarde van y</i> ✓ Both values of <i>x/beide waardes van x</i>	A A CA CA CA CA (6)

3.3.1	$r = 1.5 \text{ or } \frac{3}{2}$ $V = \frac{4}{3}\pi\left(\frac{3}{2}\right)^3$ $V = 14,14 \text{ or } \frac{9}{2}\pi$	✓ Value of /waarde van r A ✓ SF A ✓ $V = 14,14 \text{ or } \frac{9}{2}\pi$ CA
		(3)
3.3.2	$14,14 \times 1000$ $= 45000\pi \text{ or } 14137,17 \text{ l}$	✓ Method / <i>metode</i> A ✓ $14137,17 \text{ l}$ CA
		(2)
3.3.3	$\frac{14137,17}{250}$ $= 56,55$ $\approx 56 \text{ days/dae}$	✓ Method / <i>metode</i> A ✓ 56 days / <i>dae</i> A Accept / <i>aanvaar</i> 56 or/to /of/tot 57
		(2)
		[22]

QUESTION/VRAAG 4		
4.1.1	$\frac{2 - \sqrt{k-1}}{k} = 0$ $\therefore 2 - \sqrt{k-1} = 0$ $\therefore k = 5$	$\checkmark 2 - \sqrt{k-1} = 0$ A $\checkmark k = 5$ A (2)
4.1.2	$k - 1 < 0$ $k < 1$	$\checkmark \Delta < 0$ A $\checkmark k < 1$ A (2)
4.2	$3x^2 - 13x + 12 = 0$ $\Delta = b^2 - 4ac$ $\Delta = (-13)^2 - 4(3)(12)$ $\Delta = 25$ Roots are Real, Rational and Unequal. / <i>Wortels is Reëel, Rasionaal en Ongelyk</i>	$\checkmark$ Substitution / <i>vervanging</i> A $\checkmark \Delta = 25$ CA $\checkmark$ Nature of roots / <i>aard van wortels</i> CA (3)
4.3	$ax^2 - 4x + 5 = 0$ $\Delta \geq 0$ $(-4)^2 - 4(a)(5) \geq 0$ $16 - 20a \geq 0$ $a \leq \frac{4}{5}$	$\checkmark \Delta \geq 0$ A $\checkmark$ Substitution / <i>vervanging</i> CA $\checkmark$ Values of a / <i>waarde van a</i> CA (3)
		[10]

QUESTION / VRAAG 5		
5.1	$y \geq -4$	✓ Critical value / Kritieke waarde A ✓ Notation / Notasie A (2)
5.2	By inspection / deur inspeksie: $B(2; 0)$ & $C(0; -6)$	✓ $x_B = 2$ A ✓ $y_B = 0$ A ✓ $x_C = 0$ A ✓ $y_C = -6$ A (4)
5.3	$y = ax^2 - 4$ $0 = a(2)^2 - 4$ $4 = 4a$ $a = 1$	✓ Substitution / vervanging A ✓ Simplification / vereenvoudiging A ✓ Value of a / waarde van a CA (3)
5.4	$y = x^2 - 4$ $y = 3x - 6$ $x^2 - 4 = 3x - 6$ $x^2 - 3x + 2 = 0$ $(x-1)(x-2) = 0$ $x = 1$ or $x = 2$ $\therefore x = 1$ $y = 3(1) - 6$ $y = -3$ $D(1; -3)$	✓ Equating / gelykstelling A ✓ Standard form / standaardvorm CA ✓ Factors / faktore CA ✓ x value / waarde CA ✓ $y = -3$ CA (5)
5.5	$x = 0$	✓ $x = 0$ A (1)
5.6	$x \in [1; 2]$ or $1 \leq x \leq 2$	✓ Critical values / kritiese waardes A ✓ Notation / notasie A (2)
5.7	$h(x) = -(x^2 - 4) + 1$ $h(x) = -x^2 + 4 + 1$ $h(x) = -x^2 + 5$	✓ Expanding / uitbreiding A ✓ Equation / vergelyking CA (2)
		[19]

QUESTION / VRAAG 6			
6.1	$0 = \frac{2}{x} - 8$ $8x = 2$ $x = \frac{1}{4}$	$\checkmark y = 0$ $\checkmark x = \frac{1}{4}$	A CA (2)
6.2	$y = 2^0 - 8 = -7$ $0 = 2^x - 8$ $2^3 = 2^x$ $x = 3$	$\checkmark y = -7$ $\checkmark x = 0$ $\checkmark x = 3$	A A CA (3)
6.3	$x \in IR; x \neq 0$	$\checkmark x \in IR$ $\checkmark x \neq 0$	A A (2)
6.4		$\checkmark$ Asymptote / <i>asimptoot</i> For/vir $k(x)$ $\checkmark$ Both intercepts / <i>beide afsnitte</i> $\checkmark$ Shape / <i>vorm</i> For / vir $p(x)$ $\checkmark$ x -intercept/ <i>afsnit</i> $\checkmark$ Shape / <i>vorm</i>	A CA A CA A (5)
6.5	$y = -x - 8$	$\checkmark\checkmark y = -x - 8$	A (2)
6.6	$x \geq 3$ or $x \in [3; \infty)$	$\checkmark$ Critical values / <i>kritiese waardes</i> $\checkmark$ Notation / <i>notasie</i>	A A (2)
6.7	$y > -8$	$\checkmark$ Critical values / <i>kritiese waardes</i> $\checkmark$ Notation / <i>notasie</i>	A A (2)
			[18]

QUESTION / VRAAG 7			
7.1	$x^2 + y^2 = r^2$ $(3)^2 + (1)^2 = r^2$ $10 = r^2$ $r = \sqrt{10}$	✓ Substitution / <i>vervanging</i> ✓ $r = \sqrt{10}$	A CA (2)
7.2	$y = -\sqrt{10 - x^2}$	✓ $y = -\sqrt{10 - x^2}$	A (1)
7.3	$-\sqrt{10} \leq y \leq 0$	✓ Critical values / <i>kritiese waardes</i> ✓ Notation / <i>notasie</i>	A A (2)
7.4	$(0; \sqrt{10})$	✓✓ Coordinates / <i>koördinate</i>	A (2)
			[7]

QUESTION / VRAAG 8			
8.1.1	$A = P(1+i)^n$ $A = R10000 \left(1 + \frac{0,0432}{4}\right)^{4 \times 12}$ $A = R16\,746,76$	✓ Formula / formule ✓ Substitution / vervanging ✓ $A = R16\,746,76$	A A CA (3)
8.1.2	$i_{eff} = \left(1 + \frac{i}{m}\right)^m - 1$ $i_{eff} = \left(1 + \frac{0,0432}{4}\right)^4 - 1$ $i_{eff} = 4,39\%$	✓ Formula / formule ✓ Substitute / vervanging ✓ $i_{eff} = 4,39\%$	A A CA (3)
8.2.1	R 200 000	✓ R 200 000	A (1)
8.2.2	$A = P(1-in)$ $80\,000 = 200\,000(1-6i)$ $\frac{2}{5} = 1-6i$ $6i = 1 - \frac{2}{5}$ $i = 10\%$	✓ Formula / formule ✓ Substitution / vervanging ✓ $i = 10\%$	A A CA (3)
8.3	$A = P(1+i)^n$ $81 = 34(1+0,22)^n$ $(1+0,22)^n = \frac{81}{34}$ $n = \log_{(1+0,22)} \frac{81}{34}$ $n = 4,37 \text{ hours /ure}$ $n = 4 \text{ hour: /ure}$	✓ Substitution / vervanging ✓ n Subject / onderwerp ✓ $n = 4 \text{ hours / ure}$	A CA CA (3)
8.4	$A = P(1+i)^n$ $A = R50\,000 \left(1 + \frac{0,12}{2}\right)^{2 \times 3} \left(1 + \frac{0,18}{12}\right)^{12 \times 4}$ $A = R144\,935,65$	✓ $\left(1 + \frac{0,12}{2}\right)$ ✓ $n = 2 \times 3$ ✓ $\left(1 + \frac{0,18}{12}\right)$ ✓ $n = 12 \times 4$ ✓ $A = R144\,935,65$	A A A A CA
	OR / OF	OR / OF	

	$A_1 = R50\,000 \left(1 + \frac{0,12}{2}\right)^{2 \times 3}$ $A_1 = R70\,925,95561$ $A_2 = R70\,925,95561 \left(1 + \frac{0,18}{12}\right)^{12 \times 4}$ $A = R144\,935,65$	✓ Substitution / <i>vervang</i> A ✓ R70 925,95561 CA ✓ $\left(1 + \frac{0,18}{12}\right)^{12 \times 4}$ A ✓ $70\,925,95561 \left(1 + \frac{0,18}{12}\right)^{12 \times 4}$ CA ✓ R 144 935,65 CA (5)
		[18]
TOTAL/TOTAAL: 150		